FPT Education

TRƯỜNG ĐẠI HỌC FPT

Module 4 - Widgets & Basic Screens

Contents

- Introduction: Everything is a Widget
- Core Widgets (Text, Image, Icon, Card, ListTile)
- Input Widgets (Slider, Switch, Radio, Pickers)
- Layout Basics (Column, Row, ListView, Padding)
- App Structure (Scaffold, Theme, Navigation)
- Design Polish (Spacing, Consistency)
- Common Errors & Fixes
- Practice Task & Summary

Everything is a Widget

- In Flutter, the UI is built by composing widgets.
- Widgets describe structure (layout), appearance (style), and behavior (interaction).
- Every screen, every button, every text = widget

Example run with Dartpab

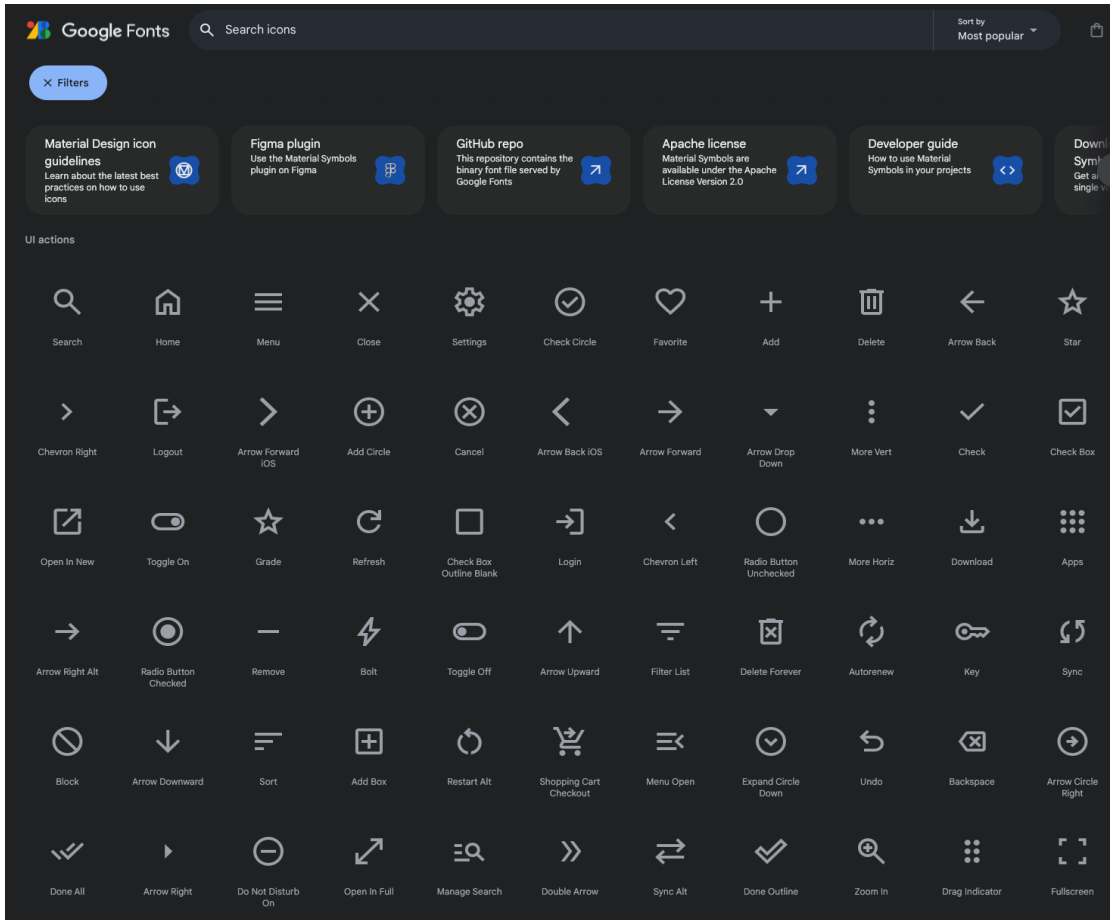
```dart
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: Scaffold(
        body: Center(
          child: Text(
            "Everything is a Widget",
            style: TextStyle(fontSize: 26, fontWeight: FontWeight.bold),
          ),
        ),
      ),
    );
  }
}
```

Reload

Output Preview: Everything is a Widget

Material Icons

- Icons in Flutter come from Material Design Icons library. They give consistent UI across devices.
- When to use
  - Navigation buttons
  - Action buttons (add, delete)
  - Status indicators



Example Material Icons

- Example: Test in DartPad: https://dartpad.dev/?id=icon-widget
- Output →

```dart
import 'package:flutter/material.dart';

void main() => runApp(const IconDemo());

class IconDemo extends StatelessWidget {
  const IconDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: Scaffold(
        body: Center(
          child: Icon(
            Icons.home,
            size: 50,
            color: Colors.blue,
          ),
        ),
      ),
    );
  }
}
```

DatePicker

- Material date picker allows users to select a date from a calendar UI
- Real uses
  - Booking apps
  - Calendar tasks
  - Form inputs (date of birth, appointment)

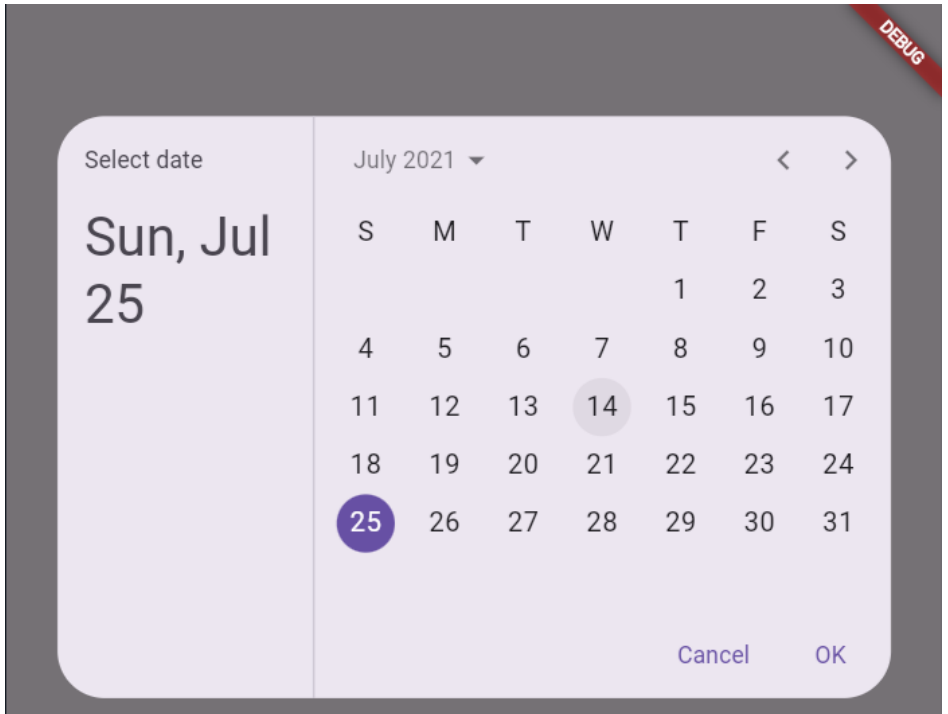


***Figure 4.2**: DatePicker*

Code Example DatePicker

```dart
import 'package:flutter/material.dart';

void main() => runApp(const DatePickerApp());

class DatePickerApp extends StatelessWidget {
  const DatePickerApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text("Date Picker Demo")),
        body: const DatePickerExample(),
      ),
    );
  }
}

class DatePickerExample extends StatefulWidget {
  const DatePickerExample({super.key});

  @override
  State<DatePickerExample> createState() => _DatePickerExampleState();
}
```

```dart
class _DatePickerExampleState extends State<DatePickerExample>
  DateTime? selectedDate;

  Future<void> pickDate() async {
    final date = await showDatePicker(
      context: context,
      initialDate: DateTime.now(),
      firstDate: DateTime(2000),
      lastDate: DateTime(2100),
    );
    if (date != null) setState(() => selectedDate = date);
  }

  @override
  Widget build(BuildContext context) {
    return Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Text(selectedDate == null
              ? "No date selected"
              : "Selected: ${selectedDate!.toLocal()}".split(' ')[0]),
          const SizedBox(height: 16),
          ElevatedButton(
            onPressed: pickDate,
            child: const Text("Pick Date"),
          )
        ],
      ),
    );
  }
```

RadioListTile (Single Choice)

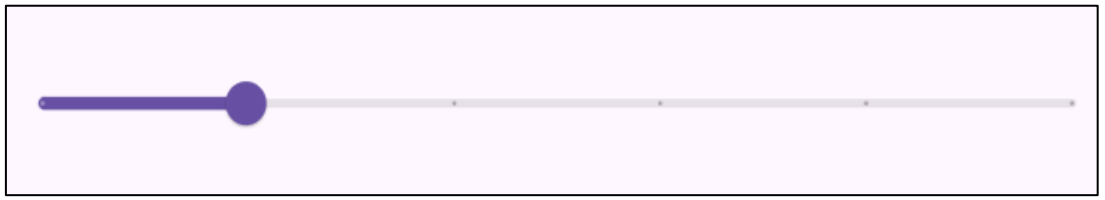
- RadioListTile lets users pick one option from a group.

- **Structure (template)**

```
RadioListTile<T>(
  title: Text('Label'),
  subtitle: Text('Optional'),
  value: /* T */,
  groupValue: /* T? */,
  onChanged: (T? v) { /* setState(() => groupValue = v); */ },
  secondary: Icon(Icons.info), // optional trailing icon
  dense: false,            // compact style
  selected: false,          // highlight when selected
)
```

Slider (Continuous Value)

- Slider allows users to adjust a value continuously between a range.

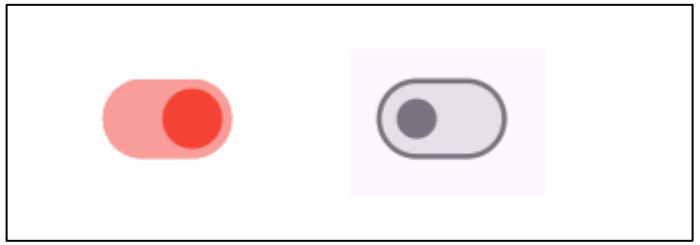


- Structure (template) →

```
Slider(
  value: /* double */,
  onChanged: (double v) { /* setState(() => value = v); */ },
  min: 0.0,
  max: 1.0,
  divisions: null, // e.g. 10 for discrete steps
  label: null,     // shows tooltip when divisions != null
  activeColor: null,
  inactiveColor: null,
)
```

Switch (Boolean Toggle)

- Switch toggles a boolean value (ON/OFF).



- Use: Enable notifications, dark mode, or toggling options.
- Structure (template)

```
Switch(
  value: /* bool */,
  onChanged: (bool b) { /* setState(() => value = b); */ },
  activeColor: null,
  activeTrackColor: null,
  inactiveThumbColor: null,
  inactiveTrackColor: null,
  thumbIcon: null, // Material 3
)
```

TimePicker (dialog)

- Concept: show TimePicker opens a Material-style time selection dialog.
- Use: Appointment scheduling, alarms, or showtime selection.

Scaffold (First Screen)

- Concept: Scaffold is the backbone of a Material screen.
- It provides AppBar, Body, FloatingActionButton, and other structural elements.
- Use: Every major screen in your app.

Theme & Dark Mode

- Concept: ThemeData defines global color, typography, and shapes across your app.
- Use: darkTheme and themeMode to switch between modes.

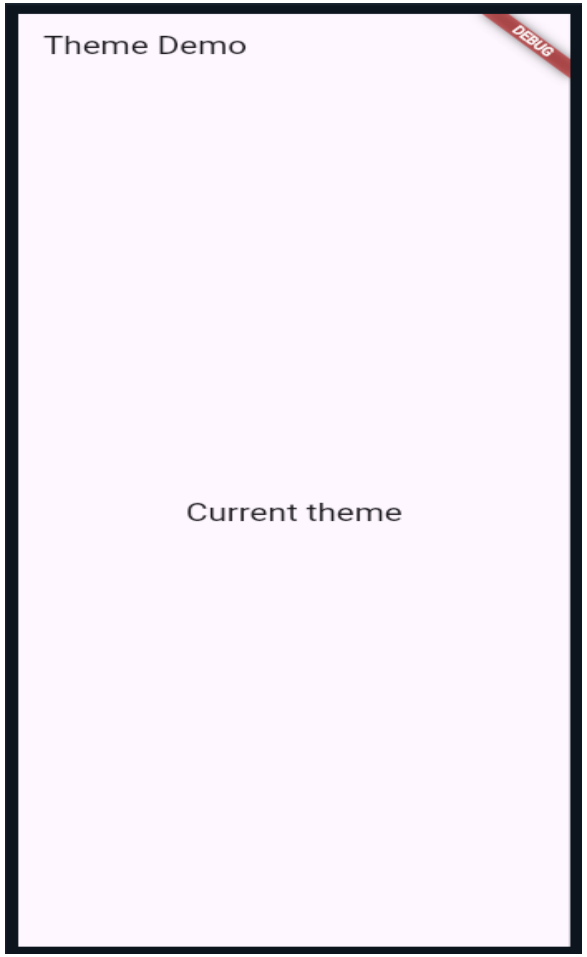
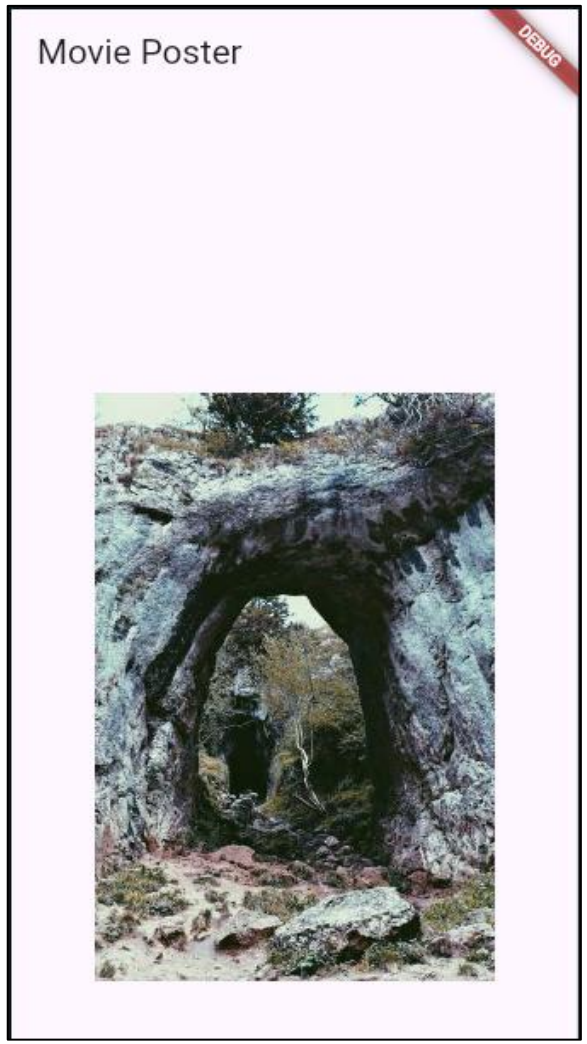


Image (Poster Detail)

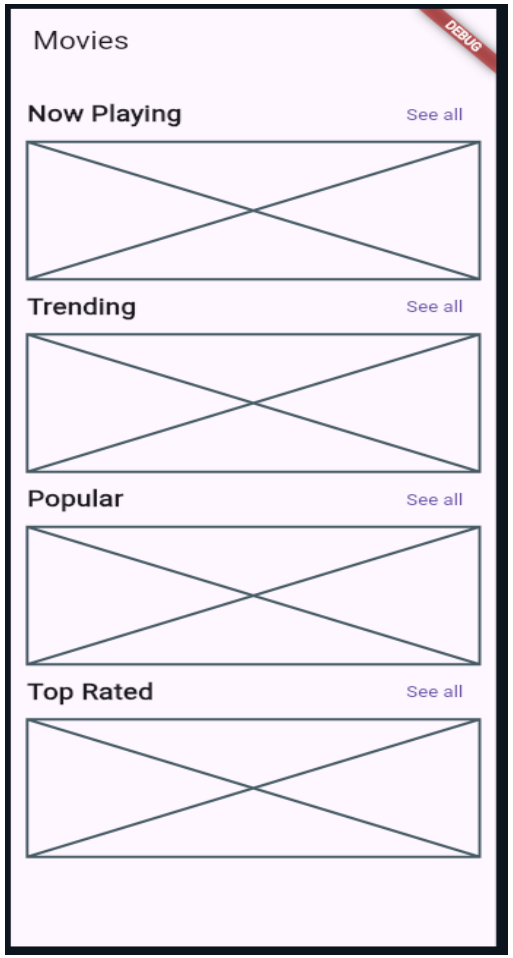
- Concept: Image.network or Image.asset displays pictures with scaling via BoxFit.
- Use Case: Used for posters, banners, or product images.

```
import 'package:flutter/material.dart';
void main() => runApp(const PosterApp());
class PosterApp extends StatelessWidget { const PosterApp({super.key});
  @override Widget build(BuildContext context) =>
      const MaterialApp(home: PosterScreen());
}
class PosterScreen extends StatelessWidget { const PosterScreen({super.key});
  @override Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Movie Poster')),
      body: Center(
        child: Image.network(
          'https://picsum.photos/400/600',
          fit: BoxFit.cover, width: 250, height: 380,
        ),
      ),
    );
  }
}
```



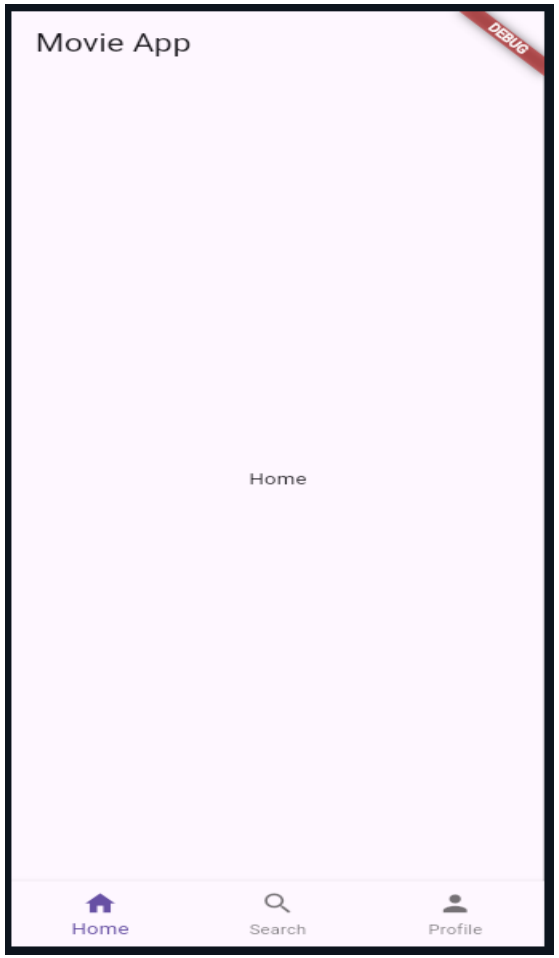
Sectioned Home Screen (Layout)

- Concept: Column, Padding, and ListView build vertical UI sections.
- Common in movie, product, or news apps.



Bottom Navigation Bar

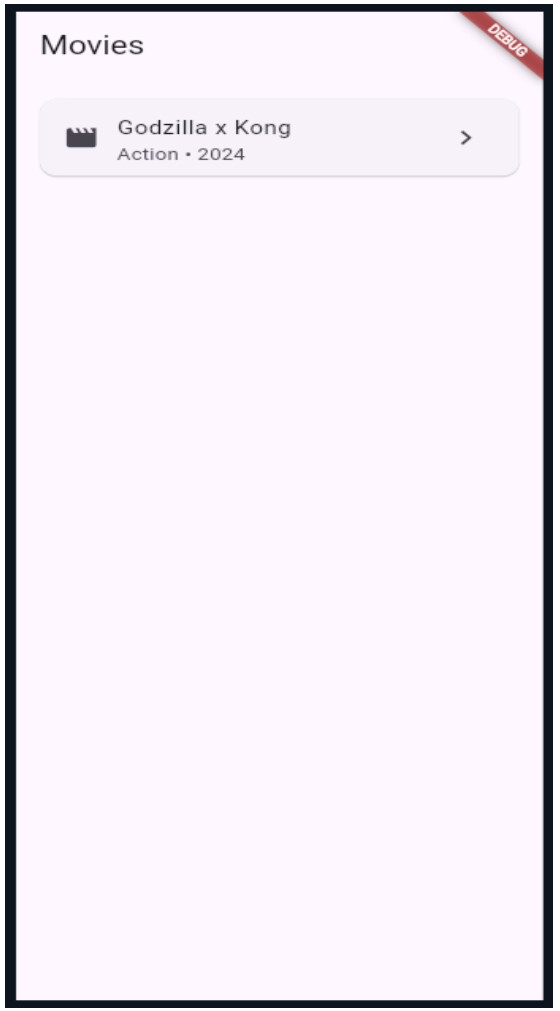
- Concept: Displays 2–5 navigation tabs at the bottom.
- Used for quick switching between app views.



Card + ListTile

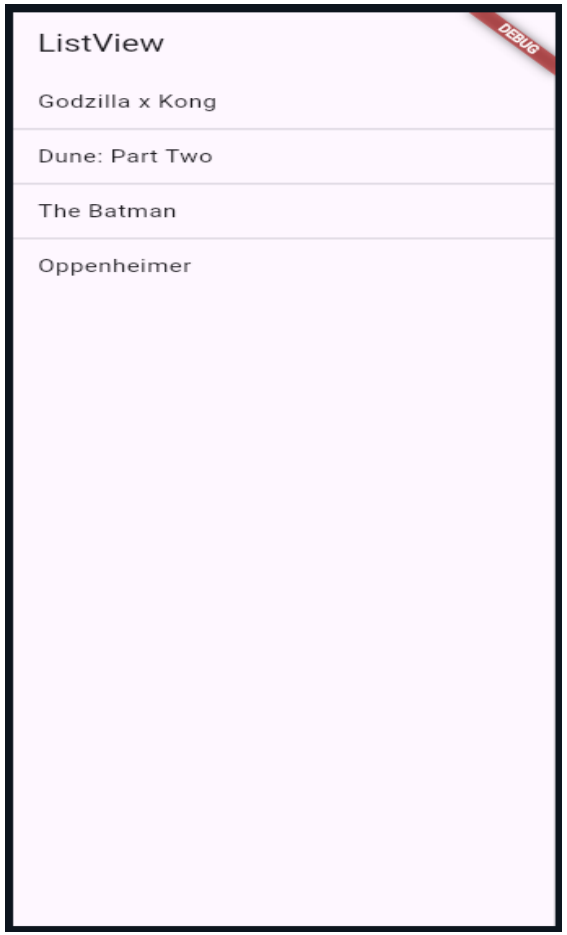
- Card + ListTile = material pattern for displaying concise information.

```
import 'package:flutter/material.dart';
void main() => runApp(const CardDemo());
class CardDemo extends StatelessWidget { const CardDemo({super.key});
  @override Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Movies')),
        body: ListView(
          padding: const EdgeInsets.all(12),
          children: const [
            Card(
              child: ListTile(
                leading: Icon(Icons.movie),
                title: Text('Godzilla x Kong'),
                subtitle: Text('Action • 2024'),
                trailing: Icon(Icons.chevron_right),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```



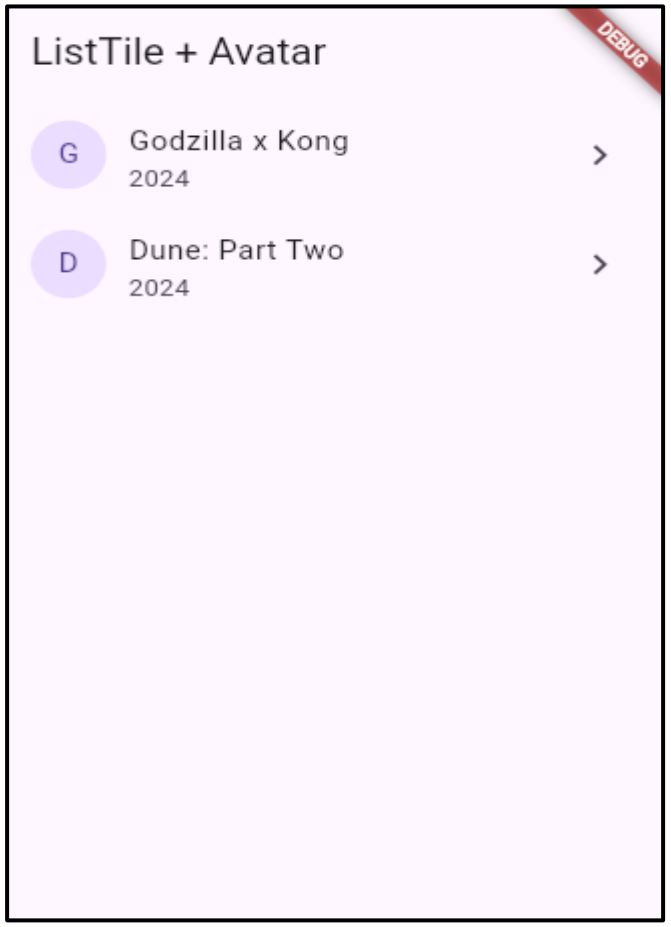
ListView Basics (Movie Titles)

- ListView creates a scrollable column; use separated or builder for long lists.
- Use Case: Movie titles, product names, chat items.



ListTile + CircleAvatar Widgets

- Concept: ListTile provides a standard row layout; CircleAvatar gives a modern leading visual.
- Use Case: Contact list, movie list, product list.

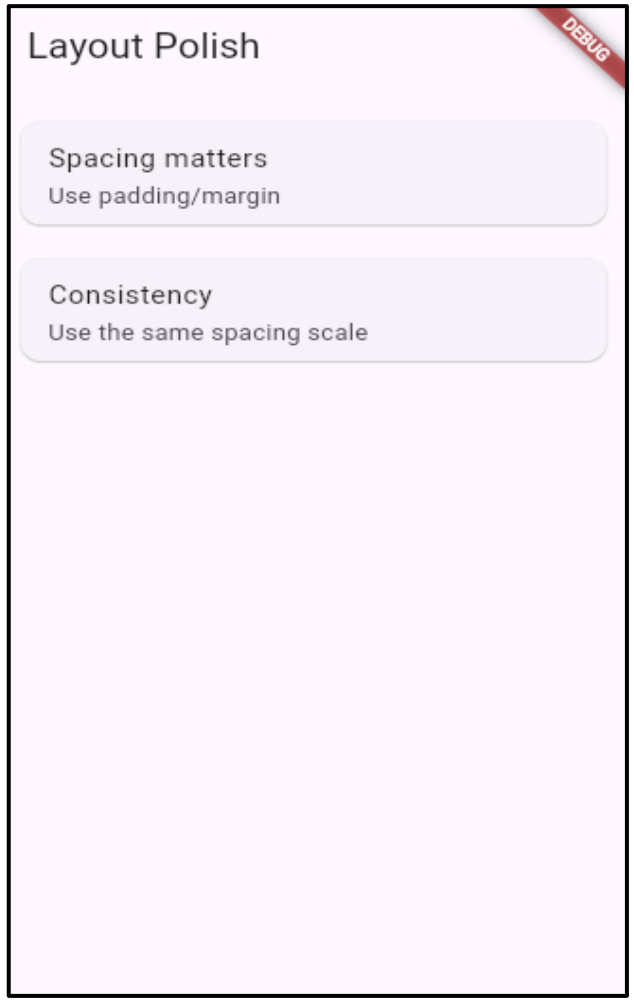


VS Code “Wrap with Widget”

- Editor-assisted refactor: wrap any widget with Padding, Center, Expanded, etc.
- Use Case: Speed up layout composition; enforce consistent spacing.

Layout Polish (Padding & SizedBox)

- Concept: Good spacing improves readability and perceived quality.
- Use Case: Apply consistent spacing scale across the app.



Date & Time Pickers (Combined)

- Concept
  Use both pickers to build booking workflows.
- Use Case
  Movie ticket booking: select both date and time

Common Errors & Fixes

Typical issues when building basic UI.

Checklist

- ListView inside Column → wrap with Expanded
- Overflow on small screens → use Flexible / SingleChildScrollView
- UI not updating → forgot setState()
- Picker call fails → make sure you call with a valid BuildContext

Practice Task

Task

- Build a movie list: title, year
- Use ListView.builder + ListTile
- onTap → show SnackBar(title)
- Bonus: replace avatar with an emoji or poster image

Summary

You Learned:

- The “Everything is a Widget” concept in Flutter
- Building screens using Scaffold, AppBar, and Body
- Using core widgets: Image, Card, ListTile, ListView
- Managing interactivity with Switch, Slider, RadioListTile, and Pickers
- Applying ThemeData and handling Dark Mode
- Improving design with Padding, SizedBox, and consistent spacing
- Identifying and fixing common layout errors

References

- Mastering Flutter 2025 – Ch.4
- docs.flutter.dev
- **https://dartpad.dev**